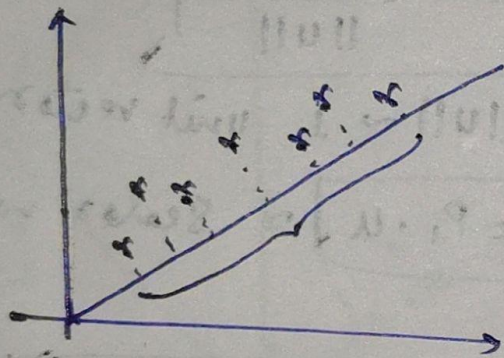


* Eigen vector and Eigen values

[Linear Transformation]

[Eigen decomposition of covariance matrix]

↓
Eigen vector &
Eigen values.



$$[A] * [v] = \lambda * v$$

↓
Eigen value

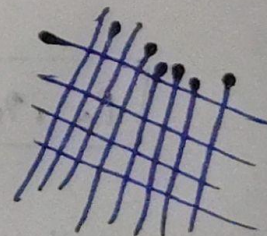
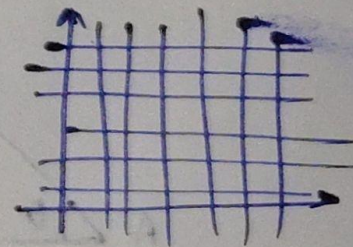
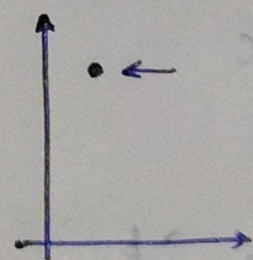
$$[A - \lambda I] * v = 0$$

↓
Eigen vector → Maximum magnitude

Eigen vector → Max Magnitude

↓
Max Eigen value

↓
Best Principal component → PC1



↓
Principal component

↓
Max Var

* Steps to calculate Eigen value and Eigen Vector

1. Covariance of features

$$\begin{matrix} \{x, y\} \\ \downarrow \\ x' \end{matrix}$$

$$\text{cov}(x, y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

$$2 \times 2 \quad \begin{matrix} & x & y \\ \begin{matrix} x \\ y \end{matrix} & \begin{bmatrix} \text{Var}(x) & \text{Cov}(x, y) \\ \text{Cov}(y, x) & \text{Var}(y) \end{bmatrix} \end{matrix}$$

$$\text{cov}(x, x) = \text{Var}(x)$$

$$\text{cov}(y, x) = \text{Var}(y)$$

$$3 \times 3 \quad \begin{matrix} & x & y & z \\ \begin{matrix} x \\ y \\ z \end{matrix} & \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \end{matrix}$$

$$\lambda_1, \lambda_2, \lambda_3$$

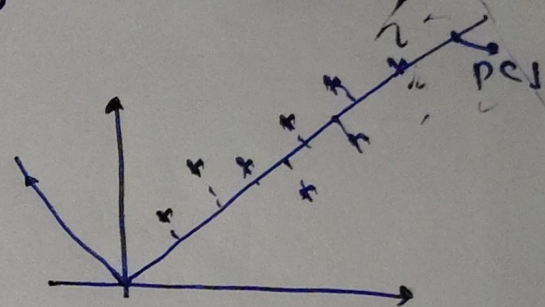
$$f_1 \neq f_2$$

$$A \cdot v = \lambda \cdot v$$

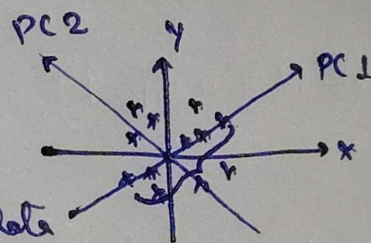
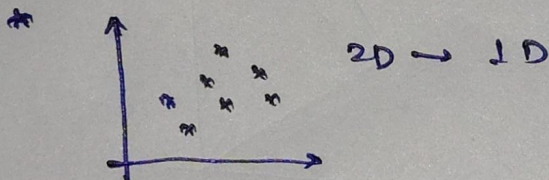
$$\begin{matrix} \lambda_1, \lambda_2 \\ \downarrow \\ \text{Eigen values} \\ \downarrow \\ \text{PC1, PC2} \end{matrix}$$

$$\begin{matrix} \lambda_1 \\ \downarrow \\ \text{PC1} \end{matrix}$$

$$\begin{matrix} \lambda_2 \\ \downarrow \\ \text{PC2} \end{matrix}$$



$\lambda_1 \Rightarrow$ magnitude of the Eigen value



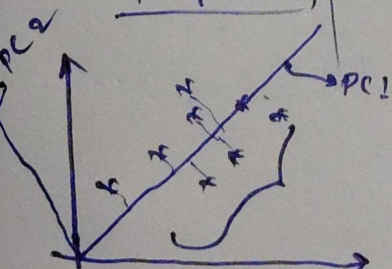
1. Standardize the data

2D $\rightarrow$ 1D

$$\begin{matrix} \lambda_1, \lambda_2 \\ \downarrow \quad \downarrow \\ \text{PC1} \quad \text{PC2} \end{matrix}$$

$$\boxed{1D}$$

Projection



2. Covariance Matrix of x & y

$$\begin{matrix} & x & y \\ \begin{matrix} x \\ y \end{matrix} & \begin{bmatrix} \text{Var}(x) & \text{Cov}(x, y) \\ \text{Cov}(y, x) & \text{Var}(y) \end{bmatrix} \end{matrix}$$

3. Find out Eigen vector and ~~vector~~ values

$$A \cdot v = \lambda v$$

$$\begin{matrix} \lambda_1, \lambda_2 \\ \downarrow \quad \downarrow \\ \text{PC1} \quad \text{PC2} \end{matrix} \Rightarrow \text{Eigen values}$$

3D $\rightarrow$ 2D

$$\begin{matrix} \lambda_1, \lambda_2, \lambda_3 \\ \downarrow \quad \downarrow \quad \downarrow \\ \text{PC1} \quad \text{PC2} \quad \text{PC3} \end{matrix}$$

$$\boxed{2D}$$

$$\boxed{1D} = 2D$$

3D $\rightarrow$ 1D

$$\lambda_1, \lambda_2, \lambda_3$$

$$\boxed{1D}$$